Chapter 9

How to use Flexible Box Layout for Responsive Web Design

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Objectives

Applied

1. Use Flexible Box Layout to lay out simple web pages or components within web pages, including the layouts for mobile devices.

Knowledge

1. In general terms, describe the use of Flexible Box Layout.
2. Describe these terms as they apply to Flexible Box Layout: flexbox, flex item, main axis, cross axis, and flex direction.
3. Explain how the flex direction affects the way these alignment properties work: justify-content and align-items.

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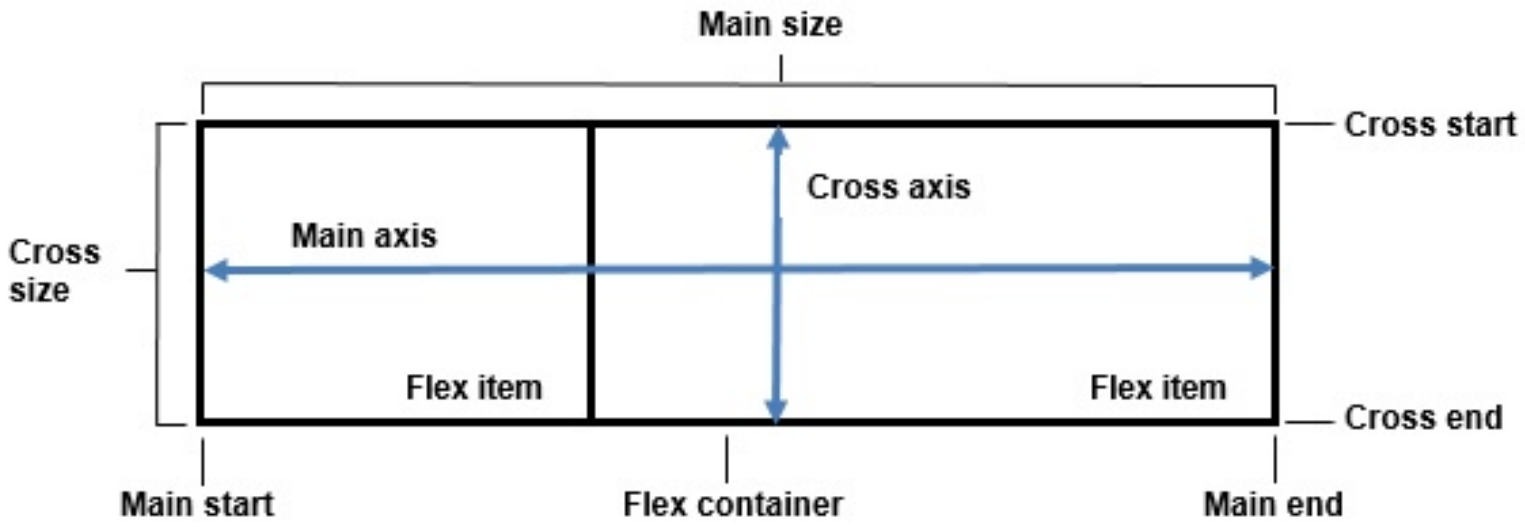
The flexible box layout



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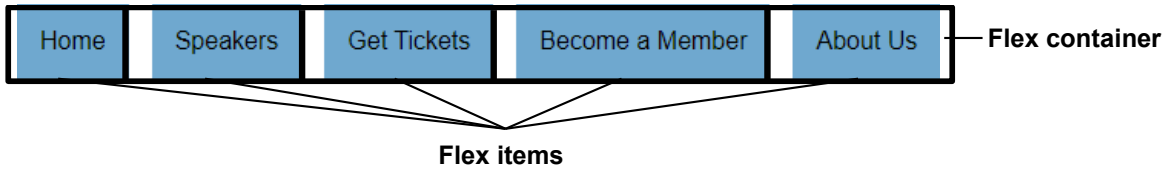
A navigation menu created using flexbox



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The HTML and CSS for a navigation menu

The HTML

```
<nav>
    <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="speakers.html">Speakers</a></li>
        <li><a href="tickets.html">Get Tickets</a></li>
        <li><a href="member.html">Become a Member</a>
        </li>
        <li><a href="aboutus.html">About Us</a></li>
    </ul>
</nav>
```

CSS that displays the menu vertically

```
ul {
    display: flex;
    flex-direction: column;
    list-style-type: none;
    padding: 0;
    margin: 0;
}
```


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The layout of the navigation menu

Home

Speakers

Get Tickets

Become a Member

About Us

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The CSS for a navigation menu with equal spacing between flex items

```
ul {
    display: flex;
    justify-content: space-between;
    list-style-type: none;
    background-color: #C8DFEE; }
li {
    background: #6FABCF;
    padding: 15px;
    text-align: center; }
a {
    text-decoration: none;
    color: #000; }
```


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The justify-content property for flex boxes

Justify-content set to space-between

Justify-content set to space-around

Justify-content set to space-evenly

Justify-content set to flex-end

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The CSS for a navigation menu that's centered vertically in a container

```
ul {
    display: flex;
    align-items: center;
    list-style-type: none;
    background: silver;
    height: 90px; }
#home { align-self: flex-start; }
```


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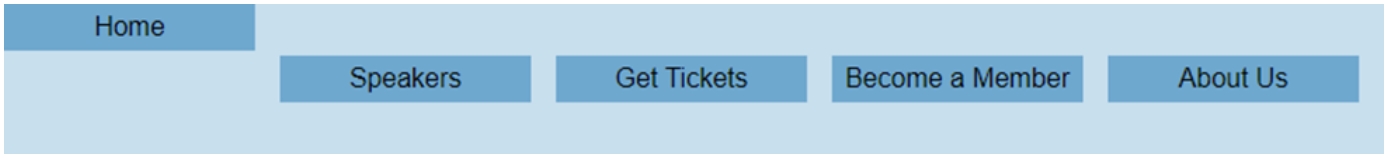
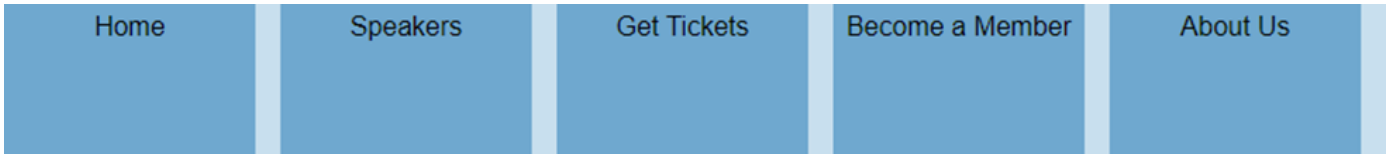
The align-items property for flex boxes

Align items is set to center



Align-items is set to stretch



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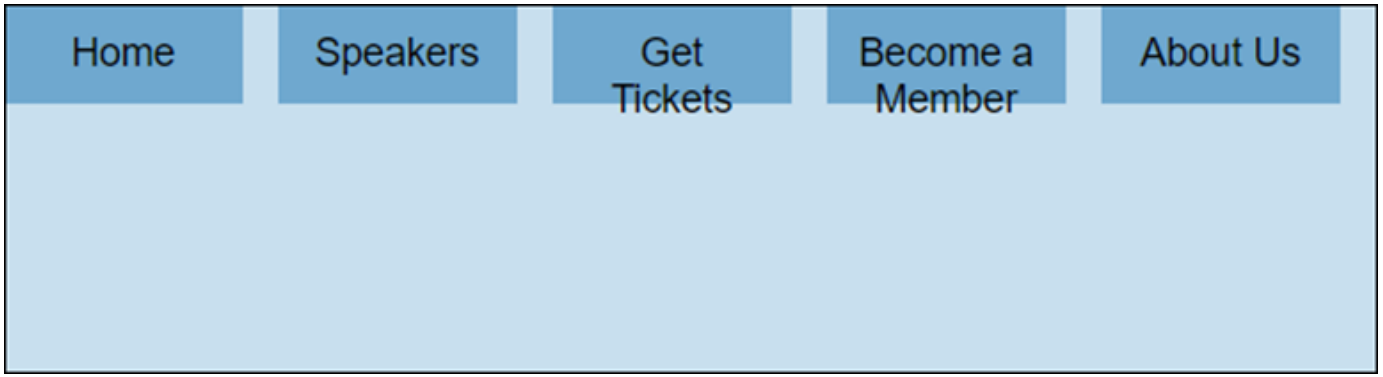
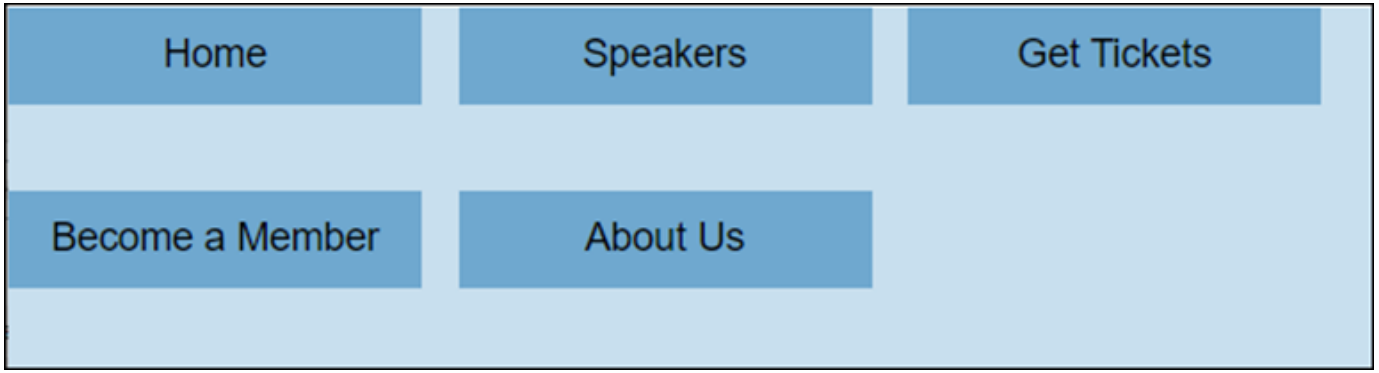
A menu with no wrapping



A menu with wrapping and default alignment

```
ul { flex-wrap: wrap; }
```



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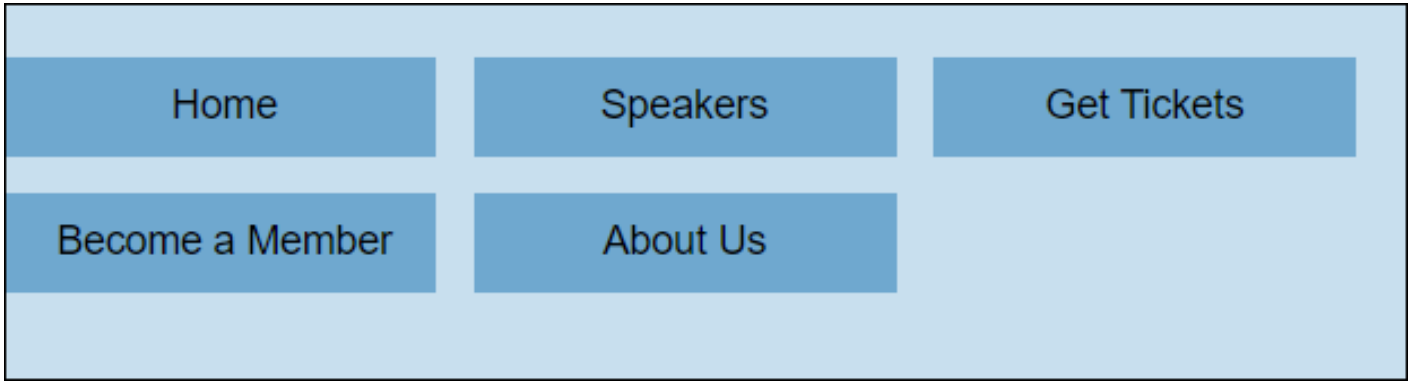
A menu with wrapping and center alignment

```
ul { flex-wrap: wrap;
     align-content: center; }
```



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The HTML for a simple page layout

```
<main>
    <nav>Navigation</nav>
    <section>Content</section>
    <aside>Sidebar</aside>
</main>
```


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CSS that uses the flex-basis property

```
main { display: flex; }
nav, aside { flex-basis: 25%; }
section { flex-basis: 50%; }
```

The layout at 400 pixels wide

The layout at 800 pixels wide

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CSS for the flex-basis and flex-grow properties

```
nav, aside {
    flex-basis: 100px;
    flex-grow: 1; }
section {
    flex-basis: 200px;
    flex-grow: 3; }
```

The layout at 400 pixels wide

The layout at 800 pixels wide

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CSS for the flex-basis and flex-shrink properties

```
nav, section, aside { flex-basis: 250px; }
nav { flex-shrink: 0; }
section, aside { flex-shrink: 1; }
```

The layout at 750 pixels wide

The layout at 500 pixels wide

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CSS that uses the flex property

```
nav, aside { flex: 1 1 100px; }
section { flex: 2 2 300px; }
```

The layout at 500 pixels wide

The layout at 750 pixels wide

The layout at 250 pixels wide

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The HTML for a page

```
<body>
    <header><h3>Header</h3></header>
    <main>
        <aside><h3>Sidebar</h3></aside>
        <section><h3>Section</h3></section>
    </main>
    <footer><h3>Footer</h3></footer>
</body>
```

A media query that changes the flex direction and reorders flex items

```
@media screen and (max-width: 760px) {
    main { flex-direction: column; }
    section { order: 1; }
    aside { order: 2; }
}
```


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The page at a narrow width with the section and aside reordered

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A speaker page in desktop layout

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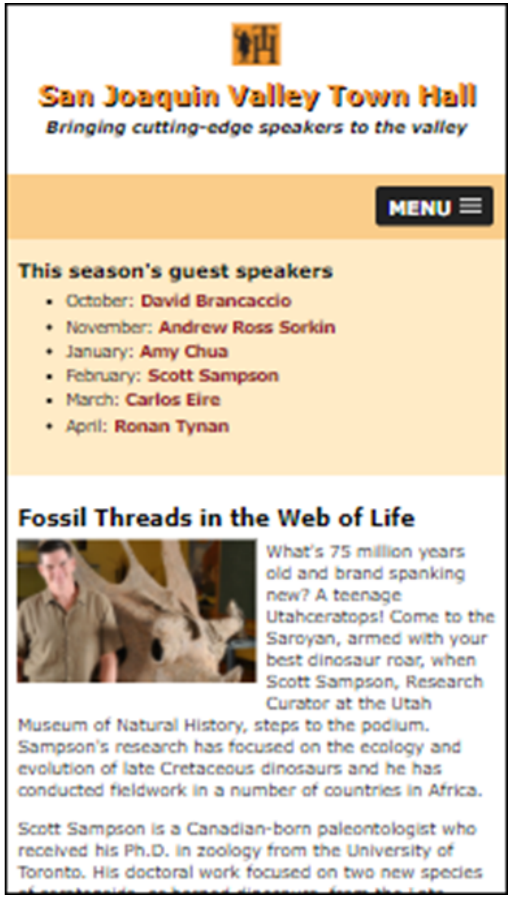
A speaker page in mobile layout



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The HTML for the structure of the page

```
<body>
    <header>...</header>
    <nav id="mobile_menu"></nav>
    <nav id="nav_menu">
        <ul>
            <li><a href="index.html">Home</a></li>
            ...
        </ul>
    </nav>
    <main>
        <article>...</article>
        <aside>
            <nav>...
                <ul>...</ul>
            </nav>
        </aside>
    </main>
    <footer>...</footer>
</body>
```


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The CSS for larger screens (part 1)

```
* the styles for the image in the header
header img {                /* image isn't scaled */
    float: left;
    margin-left: 2.0202%;
}

* the styles for the navigation menu
#nav_menu ul {
    display: flex;
    flex-basis: 100%;
    justify-content: space-around;
    list-style-type: none;
    margin: 0;
    padding: 0;
}
#nav_menu ul li {
    flex-basis: 20%;
}
```


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The CSS for larger screens (part 2)

```
/* the styles for the main element */
main {
    display: flex;
}

/* the styles for the article */
article {
    flex-basis: 60%;
    margin: 2.0202% 0 2.0202%;
    padding: 1.5em 2.52525% 0 0;
}
article img {
    margin-bottom: 1em;
    max-width: 100%;
}
```


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The CSS for larger screens (part 3)

```
/* the styles for the sidebar */
aside {
    flex-basis: 40%;
    max-height: 200px;
    min-height: 180px;
    padding: 2.0202%;
    background-color: #ffebc6;
    margin: 1.5em 2.0202% 0 0;
}
```


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The media query that changes the display to a single column

```
@media only screen and (max-width: 796px) {
    header {
        display: flex;
        flex-direction: column;
        align-items: center; }
    header h2, header h3, header img {
        margin-left: 0; }
    header img {
        float: none;
        margin-left: 0;
        flex-basis: 60px; }
    main {
        flex-direction: column; }
    article {
        order: 2; }
    aside {
        order: 1;
        margin: 0; }
}
```


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Short 9-1 Use flexbox to add a third column to the Town Hall home page